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Rotational overthrusting of the northwestern Himalaya: further palaeomagnetic evidence from the Riasi thrust sheet, Jammu foothills, India

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Thermal demagnetization results (316 samples) are presented for the Tertiary succession of the Riasi thrust sheet (Jammu foothills, northwestern Himalaya) Primary and secondary magnetization directions of Murree Group red beds (Miocene to Upper Eocene) sampled northeast of Jammu indicate, for this part of the Riasi thrust sheet, a clockwise rotation over about 45° with respect to the Indian shield since Late Eocene/Early Miocene time This accords with clockwise rotations of similar magnitude observed in the Panjal Nappe and the Krol Belt, and is interpreted as representative for the northwestern Himalaya Results from the western part of the Kalakot inlier, sampled northwest of Jammu, i e basal Murree claystone (Middle Eocene) and carbonate from the Subathu Group (lower Middle to Lower Eocene), indicate an aberrant 20–25° counterclockwise rotation which is of local importance only Available observations on rotation of Himalayan thrust sheets with respect to the Indian shield, indicate that the Himalayan Arc has formed through oroclinal bending This supports Powell and Conaghan's and Veevers et al 's model of Greater India with large-scale intracontinental underthrusting along the Main Central Thrust beneath the Tibetan Plateau Minimal magnitudes of underthrusting of 550 km in the Krol Belt and 650 km in the Thakkhola region are concluded Palaeolatitude observations (herein and in [1[) agree with absolute positioning of the Indian plate based on India-Africa relative movement data fixed to a hotspot frame in the Atlantic Ocean, and with palaeolatitude observations from DSDP cores on the Indian plate Collision-related secondary magnetic components observed both to the north and to the south of the Indus-Tsangpo Suture zone show palaeolatitudes between the equator and 7°N Comparison of both datasets indicates that initial contact between Greater India and south-central Asia had been established in the Hindu Kush–Karakorum region by about 60 Ma ago whereas eastwards progressive suturing had advanced to the Lhasa Block segment of the Indus-Tsangpo Suture zone before 50 Ma ago

1. Introduction

Seafloor spreading [2–6] and palaeomagnetic data [7–11] have established Greater India's Early Cretaceous breakaway from Eastern Gondwana and subsequent northwards flight till complete suturing with south-central Asia at about 55 Ma ago. This occurred at equatorial palaeolatitudes as indicated by suturing related secondary magnetic components [9] observed both to the north and to the south of the Indus-Tsangpo Suture zone (ITS). Widely divergent hypotheses have been postulated for the subsequent post-collisional deformation. Interpretations for Greater India's large-scale impingement into south-central Asia over about 30° of latitude from the equator to its present 30°N position in Asia, vary widely from Dewey and Burke's [12] basement reactivation model and Powell and Conaghan's [13] and Veevers et al 's [14] intracontinental underthrusting model to the indentation [15] and propagating extrusion tectonics [16] model of Tapponnier and coworkers See [17–19] for discussion of models.

The recent French-Chinese cooperative field

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program in Tibet [20,21] has not shown conclusive evidence for the basement reactivation model, and the attractive extrusion model ignores the problem of the double thickness of the Tibetan crust Powell and Conaghan [13,17] interpreted this double thickness in terms of large-scale intracontinental underthrusting of Greater India along the Main Central Thrust (MCT) beneath the Tibetan Plateau, may be as far north as the Kun Lun–Astin Tagh–Nan Shan region. This finds support in several recent analyses of surface- and body-wave propagation beneath the Tibetan Plateau and the Himalayan Arc [22–25]. Deep seismic sounding studies [26] have shown the presence, albeit only locally established sofar, of a double Moho reflection at depths of about 50 and 70 km

Seeber et al. [27], followed by Ni and Barazangi [28], have interpreted seismic data from the Himalayan and Tibetan regions in terms of a model for intra-continental underthrusting, which is an outgrowth of Powell and Conaghan's [13] "steady-state" model They draw attention to the surprisingly circular outline in plan within the central region of the Himalayan Arc, and less so close to the eastern and western syntaxes, of the outline of the MCT, the southern topographical edge of the Tibetan slab, and a closely associated zone of frequent seismic activity of intermediate magnitude attributed to intracontinental underthrusting They reach the important conclusion that such a neatly circular outline cannot be accidental and must be of secondary origin, representing a now stabilized state of continental underthrusting comparable to oceanic subduction along arcuate island arcs Such an analogy was stressed before by Molnar and coworkers [29,30] from gravity and geometric observations.

This hypothesis can be tested palaeomagnetically, as it predicts along the Himalayan Arc a gradually changing pattern of rotations In the following we present new palaeomagnetic results from the Jammu region which together with data from the wider Himalayan region [19] confirm a young secondary origin for the Himalaya Arc, and support earlier advanced arguments for an equatorial palaeolatitude of collision between Greater India and south-central Asia

2. Geological setting

Structural trends in the Western Himalayan Syntaxis change from NNW-SSE and NW-SE in the eastern limb of the syntaxis, to N-S and NE-SW in the western limb of the syntaxis (Fig. 1 [31,32]) In the outermost thrust zones this bend becomes acute and the structural trend turns nearly upon itself from NNW-SSE to NNE-SSW to continue into a WSW trend in the Eastern Salt Range. This acute reentrant is of superficial nature only Its development is facilitated by detachment in the Potwar Plateau–Salt Range region of the suprastructure along Eocambrian evaporites of the Salt Range Formation (or Saline Series, [33]). The MBT and the MCT which approach each other in outcrop in the eastern limb of the syntaxis, known there as Murree Thrust and Panjal Thrust respectively, continue with a northwestern strike in the basement northwest of the syntaxis as the Indus Kohistan Seismic Zone (IKSZ [34])

The Kahsmir Basin has been thrusted southwestwards as part of the Panjal Nappe over the Panjal Thrust which surfaces at the foot of the Pir Panjal Range in the Jammu-Kashmir region A palaeomagnetic reconnaissance study of the Kashmir Basin [35] has shown that this overthrusting was characterized by a Late Tertiary clockwise rotation over about 45° relative to the Indian shield The thrust-slice between the Panjal Thrust and the directly southwards adjacent Murree Thrust is made up of a sequence of Late Palaeozoic Panjal Traps overlain by carbonate of the Subathu Group [36,37]

Further south within the Jammu foothills, the Riasi Thrust (Fig 1) forms a major dislocation zone, along which the Murree Group and locally basement and overlying Subathu Group sediments have been overthrust to the south onto Siwalik Group sediments. The Riasi-Kalakot basement uplift north of the Riasi Thrust gives a good overview of the regional stratigraphy The core of the inlier is made up of the Sirban (Great or Jammu) Limestone of Late Precambrian to Cambrian age [38], unconformably overlain by carbonate from the Subathu Group and molasse deposits of the Murree and Siwalik Group (Fig 1 [39,40])

The stratigraphic succession of the Subathu Group in the Kalakot inlier (Fig. 2) starts with

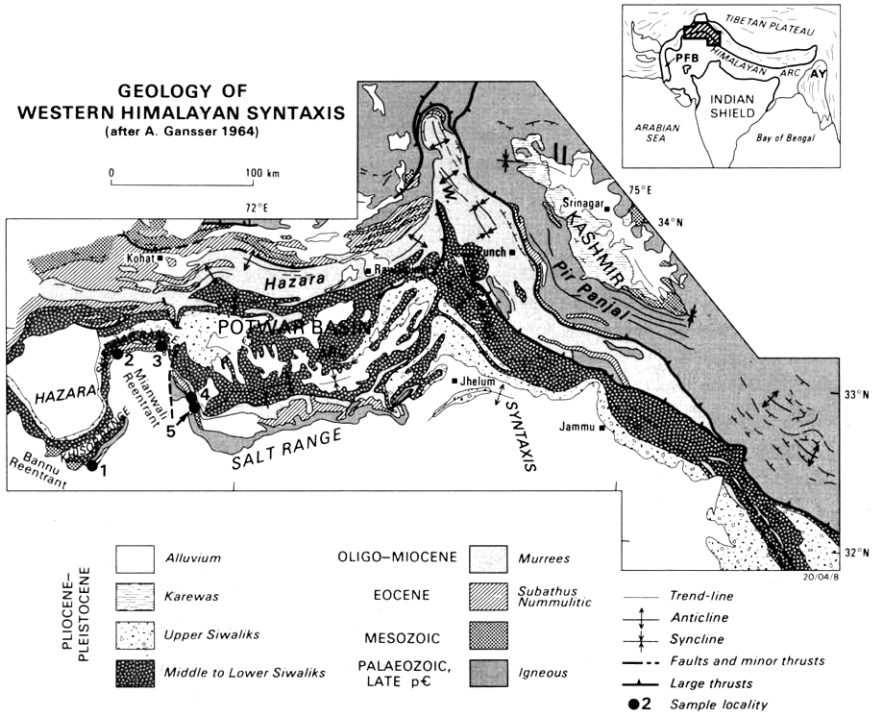


Fig 1 Geological map of the Western Himalayan Syntaxis after Gansser [63] Sampling sites are indicated by full dots See Table 1

chert breccia and bauxite of the Jangalgali Formation, overlain by the Beragua Formation made up of sandstone, carbonaceous shale with coal seems and ferruginous shale The succeeding Kalakot Formation is characterized by olive-grey to khaki needle shale interbedded with dark grey fossiliferous limestone and argillaceous limestone. The topmost Arnas Limestone is made up of grey thick-bedded fossiliferous limestone, shelly limestone and greenish unfossiliferous limestone, overlain with a gradational contact by ossiferous purple claystone of Middle Eocene [39,40] or possibly early Late Eocene age [37,41,42] which forms the base of the Murree Group.

The Murree sequence in the Jammu region is thought to have developed without a distinct sedimentational break, though detailed biostratigraphical control is not available for the major part of the Murree Group sequence. Lithologically the Murree Group is subdivided into the Lower Murree, mainly argillaceous with thick sections of deep purple to chocolate coloured shale and siltstone, and the Upper Murree, more arenaceous and purple coloured with clay and siltstone beds of a predominantly purple and subordinately greenish-grey colour. The Siwalik Group of Middle Miocene to Pleistocene age overlies the Murree Group, and is divided on lithological grounds [33] into Lower, Middle and Upper Siwaliks. The Lower Siwaliks, like the Upper Murrees, consist of an alternation of fine- to medium-grained grey coloured sandstone and reddish-brown clay and siltstone.

3. Sampling

Sampling was carried out over selected stratigraphic profiles in two regions of different regional strike. The Chinji Formation [33] of the Lower Siwaliks and two sections in the Murree Group were sampled at three localities along the Jammu-Srinagar national highway (Fig. 1, Table 1). The predominant strike in this region conforms with the NW-SE trend characteristic for the northwestern Himalaya. The basalmost Murree beds and the upper part of the Subathu Group were sampled at two localities in the Kalakot inlier (Fig. 1, Table 1), where the structural trend deviates from the overall northwestern Himalayan trend into an east-west direction.

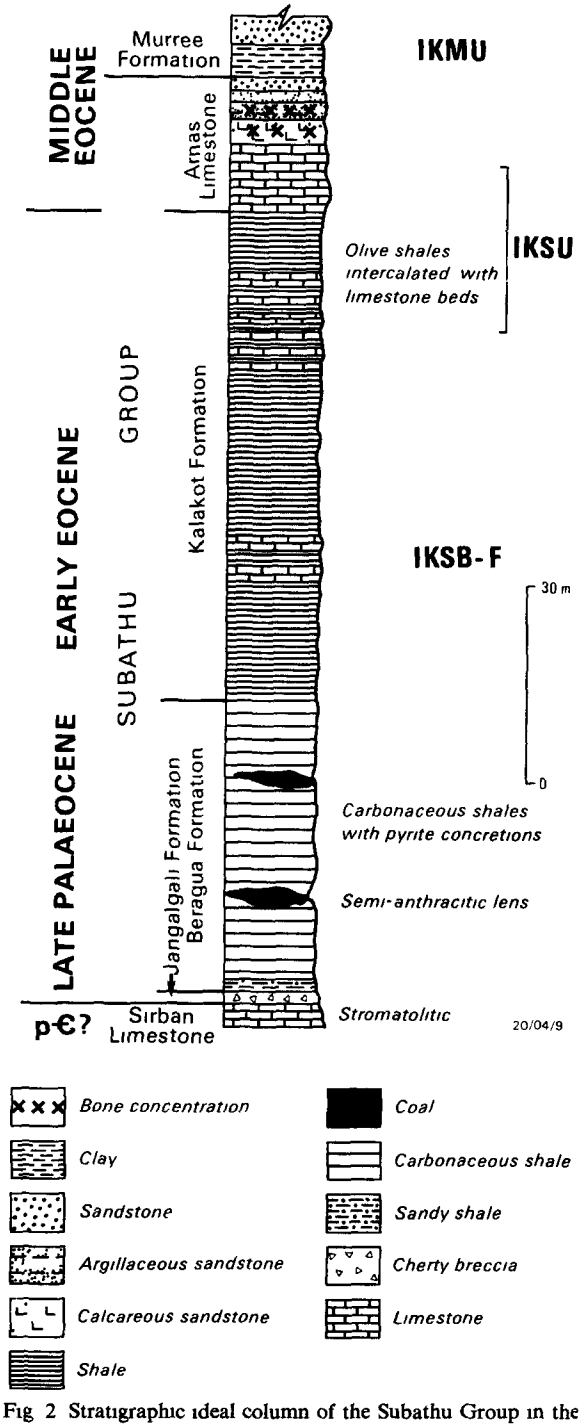


Fig 2 Stratigraphic ideal column of the Subathu Group in the Kalakot inlier after Sahni and Khare [40] and Singh [39].

—The Chinji Formation (Fig. 1, location 1) was sampled over about 300 m stratigraphically, mainly directly north of the highway tunnel. From 6 sites (ICNA–E, ICNG) a total of 68 core samples was taken throughout grey-greenish medium grained impure and very thick-bedded sandstone of floodsheet origin, moderately to steeply dipping towards the southwest (135–150°/44–67°SW). Intercalated red-brown clay beds were too fragmented to be drilled.

—Further north along the national highway, about 15 km south of Kud (Fig 1, location 2), 53 samples (25 core- and 28 block-samples, IBMU1–28) were taken over a stratigraphic interval of 65 m in a succession of brown-purple coloured thick-bedded sandstone grading upwards into silt and clay, which form part of the moderately west dipping northern limb (335–15°/41–44°W) of an anticlinal structure. The stratigraphic position of the sampled beds within the Murree Group could not be established, though unidentified plant beds suggested that the succession might belong to the "lower upper" Murree. This has not been substantiated from our palaeomagnetic results

—About 2 km north of Batote (Fig. 1, location 3) along the highway, another 60 core samples (IBML1–60) were taken over about 62 m stratigraphically in brown-purple coloured siltstone and claystone beds alternating with very thick-bedded sandstone, dipping moderately towards south-southwest (100–110°/30–37°SSW). The stratigraphic position of the sampled succession is equally uncertain. A "lower" Murree affinity suggested on lithological appearance, is contradicted by our palaeomagnetic results.

—In the Kalakot inlier along the east bank of the Tawi river about 1 km southwest of Siar (Fig. 1, location 2, Fig. 2), basalt beds of the Murree Group and the upper part of the Subathu Group were sampled. The river profile shows local tectonic disturbance which were avoided in the sampling. A total of 28 samples (5 core-, 23-block-samples, IKMU1–28) were taken over a stratigraphic interval of 3 m in purple claystone at the base of the Murree Group which dips slightly to the south (82°/18.5°S). Another 52 core samples (IKSU1–52) were taken over about 20 m stratigraphically in the immediately underlying upper part of the Kalakot Group, i.e. in the fossiliferous grey thick-bedded Arnas Limestone (Ans Limestone member) and in thin-bedded limestone of the Kalakot Formation interbedded in a pre-

TABLE 1

Sampling details

Formation, identification	Locality (°N, °E) (Fig 1)	Age	Lithology	Section length (m)	Samples (spec)	Initial intensity (mA m^{-1})	Dip direction (°)	Dip (°)
Lower Siwaliks Chinji Formation, ICNA-E, ICNG	32 85, 74 92 (1)	Late Miocene	red clays (ICNB), greenish grey sst (ICNA, C-E, G)	300	68 (127)	0 199–1 862	135–150	44–67 SW
"lower upper" Murree, IBMU	33 12, 74 30 (2)	Eocene/Oligocene probably	brown purple thickbedded sst, siltst, clay	65	53 (222)	0 283–6 463	335–15	41–44 W
"lower" Murree, IBML	33 12, 74 30 (3)	Oligocene/Miocene probably	brown purple siltst, clay thickbedded sst	62	60 (107)	0 309–5 650	100–110	30–37 SSW
basal Murree, IKMU	34 34, 73 85 (4)	Middle Eocene	purple claystone	3	28 (44)	0 402–8 990	82	18 5 S
Arnas Limestone, top Kalakot Fm, IKSU	34 34, 73 85 (4)	Early/early Middle Eocene	thickbedded lst (Arnas), thinbedded lst in shaly seq (Kalakot)	20	52 (95)	0 199–1 786	95–105	19–30 S
basal Kalakot Fm, IKSB-F	34 34, 73 85 (5)	Early Eocene	argil lst and massive lst	45	55 (87)	0 114–2 314	90–147	14–20 SSW

dominantly shaly sequence. The sampled beds have a slight to moderate dip towards SSW (95–105°/19–30°SSW).

—In a section along the Siar-Sialsui road about 3 km northwest of Siar (Fig. 1, location 5, Fig. 2), and in the nearby Nala above the Beragua Mine, a

TABLE 2

Summary of palaeomagnetic directions and pole positions

No	Component [a]	Mean specimen direction						South pole position				
		D (°)	*I* (°)	*k* (°)	α95	*N* (N/R)	polarity (°S)	lat (°E)	long (°)	*dp* (°)	*dm* (°S)	palaeolat
A Chinji Formation, ICNA–E, G, recent field components only												
1	Rn ICNG		352	45 5	55	3 5	29	N				
2	Rn ICNE	357	44	70	4	21	N					
3	Rn ICND	352	49	88	2 5	32	N					
4	Rn ICNC	353	39	27	7 5	15	N					
5	Rn ICNB	356 5	42	21 5	8	17	N					
6	Rn ICNA	346 3	49 5	46	4	26	N					
B "lower upper" Murree, IBMU												
7	Rn	342	52	13 5	7	32	N					
8	Sr	195 5	−36 5	62	5	14	R	71	23 5	3 5	6	20
9	Pr	204 5	−23	20	3 5	91	R	59	22	2	3 5	12
C "lower" Murree, IBML												
10	Rn	1 5	52 5	33	3	75	N					
11	(Con	18	66	16	4 5	60	N)					
12	(Cor	201	−16 5	10	9	28	R)					
13	Pr	26	38 5	9 5	7	51	N	64 5	4 5	5	8	21 5
14	Pn	200 5	−40	24	5	35	R	69	8	3 5	6	22 5
15	Prn (comb)	23 5	39	12 5	4 5	86	51/35	66 5	6	3	5 5	22
D basal Murree, IKMU												
16	Rn	6 5	52	13 5	15 5	8	N					
17	Sr	143 5	−41	34 5	7	14	R	56 5	155	5	8 5	23 5
18	P1r	129	−10	26 5	4 5	38	R	34 5	144	2 5	4 5	5
19	P2r	127 5	−15	45	3 5	41	R	35 0	147 5	2	3 5	7 5
20	P12r	129	−12 5	40	3 5	39	R	35	145	2	4	6 5
E Subathu Group, Arnas Limestone and upper Kalakot Formation, IKSU												
21	Rn soft	357	38 5	52 5	2 5	74	N					
22	Rn hard	323	39	14	8 5	23	N					
23	Rr [b]	168 5	−51	43	2 5	66	R	80	152 5	2 5	3 5	32
24	Sr? [c]	141 5	−28	25	11 5	8	2/6	50	144	6 5	12 5	15
F Subathu Group, basal Kalakot Formation, IKSB–IKSE												
25	Rn IKSC	338 5	49 5	16	17	6	N					
26	Rn IKSD	354 5	42	17 5	12 5	9	N					
27	Rn IKSE	338	46	24	16	5	N					
28	Rn IKSF	351 5	45 5	28 5	7 5	14	N					
29	Rn IKSF hard	343	54	34 5	9	9	N					
30	Sr IKSF	182	−35	18 5	12	9	R	75	66	8	14	19
31	Pr IKSE	129	2	29	8 5	11	R	31	138 5	4 5	8 5	1 (N)

D = declination, I = inclination, k = precision parameter [66], α_{95} = halfangle of cone of 95% confidence, N = number of specimens, dp, dm = half angles of the oval of 95% confidence about the pole position

[a] R = recent field component (Recent field direction in Jammu region $D = 1°$, $I = 49\,5°$), S = secondary component, P = primary component, Co = artefact resulting from composite breakdown of a multiple component system, n = normal polarity, r = reversed polarity

[b] Recent field component of reversed polarity and of pre-folding origin

[c] Probably of secondary origin, see text

total of 55 samples (45 core-, 10 block-samples) was taken from 5 sites (IKSB–F) over about 45 m stratigraphically in the more basal part of the Kalakot Formation, which is made up of thin-bedded argillaceous limestone and more massive limestone interbedded with olive-coloured shale, dipping slightly to the south and southwest (90–147°/14–20°SSW) at the sampled locality.

Samples were oriented in the field with both a magnetic and a solar compass.

4. Analysis and results

All sampled cores and cored block-samples (316) were sliced into specimens (682) of 2.25 cm height and 2.5 cm diameter and were washed in diluted HCl prior to measurement of initial remanence. All measurements were carried out on a two-axis ScT cryogenic magnetometer interfaced with a HP 2100 minicomputer, programmed for on-line data reduction. Selected pilot specimens, generally one specimen every third sample throughout a profile, were progressively thermally demagnetized in 13–16 steps up to 500–600°C for the Subathu limestone samples, and in 16–21 steps up to 685°C for the red beds and sandstone from the Murree and Siwalik Group. The remainder of the specimens were partially progressively demagnetized in 11–14 steps between 110° and 500°C for the limestone samples and in 15–22 steps between 110° and 675°C for the red beds and sandstone samples.

All heatings were carried out in large-volume furnaces with a feed-back controlled 10-set Helmholtz coil system [43] reducing the ambient field to less than 5 nT over the furnace space. Heating (1–5/4 hr) was done in an inert argon gas environment, with forced cooling in air (1/4–3/4 hr). Onset of significant chemical changes and in particular formation of viscous magnetization phases was monitored through bulk susceptibility measurements after each heating step.

Identification of individual magnetic components was based on visual inspection of Zijderveld plots [44], and directions were determined with an adapted version of Kirschvink's [45] principal component analysis program. Density distributions of all analyzed component directions, both corrected and non-corrected for bedding, were scrutinized for groupings, on basis of which mean component directions were computed and interpreted.

With exception of the Siwalik sediments, the rocks showed general presence of three magnetic components whose interpretation will be discussed in the following chapter.

(1) A generally soft recent field component of normal polarity and post-folding origin. A soft component of reversed polarity and of recent but pre-folding origin was present also in the upper part of the Subathu Group succession.

(2) A component of intermediate blocking temperatures, attributed to a Middle Tertiary thermochemical event associated with formation of the Main Central Thrust. In one of the Murree Group successions such a component could be identified as an artefact resulting from overlapping blocking temperature spectra of a hard (3) and a soft (1) component.

(3) A hard component of predominantly normal polarity, interpreted as a primary magnetization.

Siwalik sandstone. Intensity of initial remanence (NRM) ranged between 0.2 and 1.9 mA m^{-1}. The specimens showed consistent and predominant presence of a soft normal polarity component of large intensity (60–90% of initial NRM), which was eliminated at temperatures up to 250–300°C (Table 2: 1–6). Its direction close to the local present field identifies this magnetization as a secondary component of post-tectonic and obviously recent origin. Demagnetization at higher temperatures failed to show any consistent direction of pre-tectonic origin.

Murree red beds, Kud and Batote Intensity of initial remanence ranged between 0.3 and 6.5 mA m^{-1}. Magnetic response to thermal demagnetization was uniform in all specimens. Specimens from both the "lower upper" (Kud) and the "lower" (Batote) Murree sections showed consistent presence of three magnetic component (Figs. 3A–G, 4A–D, Table 2: 7–15):

(1) A soft component representing about 15–35% of initial NRM, which was generally removed at 250–350°C.

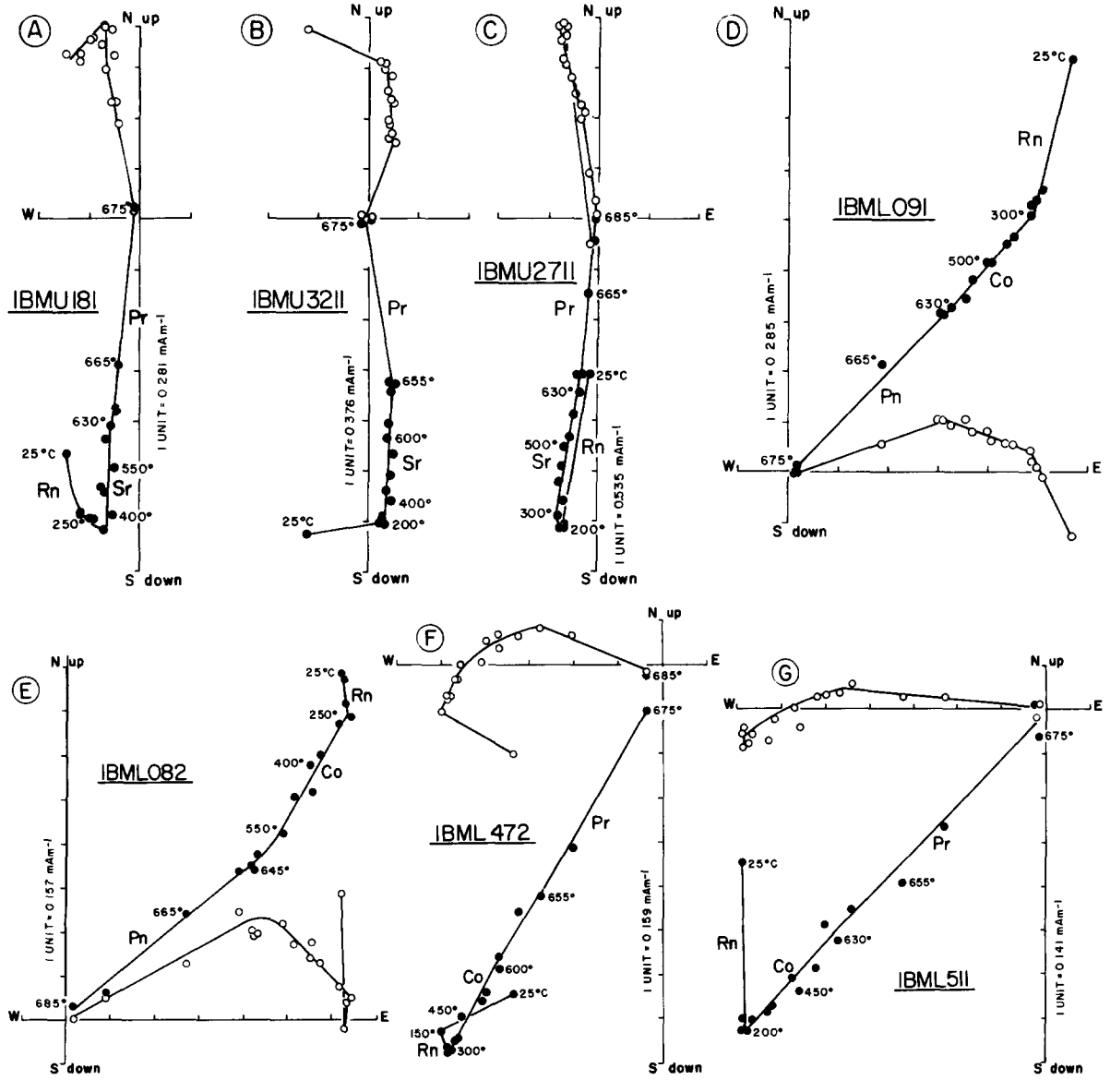
(2) An intermediate component representing about 20–40% of initial NRM, gradually broken down in the succeeding interval up to 550–600°C

(3) A hard component representing about 40–50% of initial NRM, removed at 675–685°C with a pronounced blocking temperature range between 655° and 675°C (Fig. 4A–D).

The soft component was exclusively of normal polarity (Fig. 3A–G). The intermediate and hard components were of normal polarity in the "lower upper" Murree section (Fig. 3A–C) and in the bottom part of the "lower" Murree section (IBML1–29, Fig. 3D, E), but changed both at the same level to a reversed polarity and remained so throughout the upper part (IBML30–60, Fig. 3F, G) of the "lower" Murree section

Basal Murree red claystone, Kalakot Initial NRM intensity ranged between 0.4 and 9 mA m^{-1} All red claystone specimens showed a very consistent and uniform demagnetization pattern with presence of four well-determinable magnetic components (Figs 3H–J, 4E, Table 2 16–20):

(1) A soft component of low intensity with dispersed directions and probably of viscous



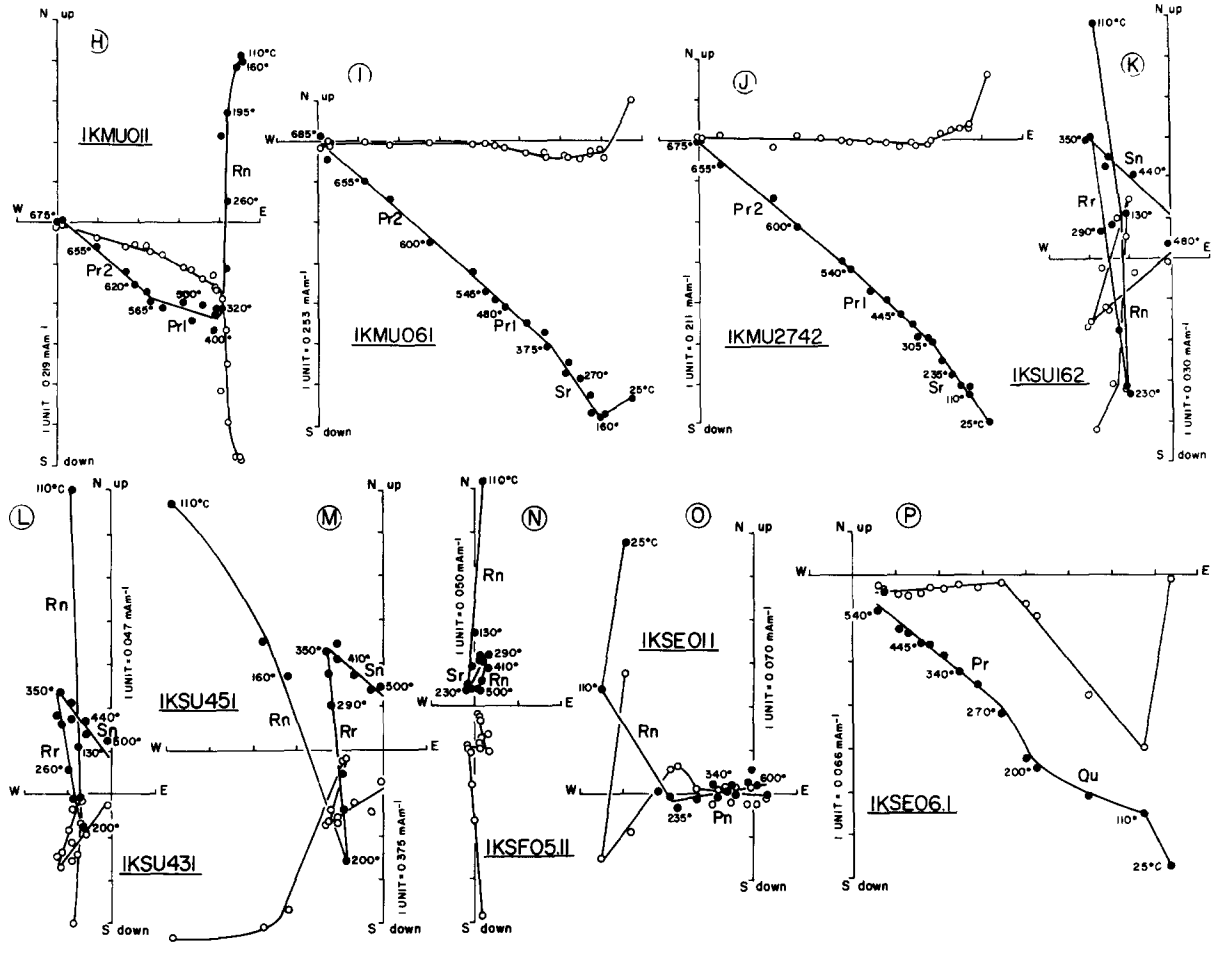


Fig 3 Zijderveld [44] diagrams of some representative specimens during thermal demagnetization The circles indicate successive positions—in orthogonal projection—of the end points of the resultant magnetization vector during progressive thermal demagnetization Open circles indicate projections on the vertical east-west plane, and dots indicate projections on the horizontal plane Numbers denote successive peak temperature values R = recent field component, S = secondary component, P = primary component (with subscript 1 or 2, see text and Table 2), Co = composite breakdown of a multicomponent system, n = normal polarity, r = reversed polarity Results are not corrected for bedding

origin, representing 10–15% of initial NRM, was eliminated at 100–160°C (Fig. 3I, J).

(2) An intermediate component representing about 15–20% of initial NRM was eliminated at temperatures up to 300–400°C. This component was of reversed polarity (Fig. 3I, J), except in the five cored samples (IKMU1–5) where overheating during an abandoned drilling procedure may have introduced a normal polarity component of recent origin (Fig. 3H)

(3) and (4) In the succeeding temperature interval up to 685°C two magnetic phases of comparable (Fig. 3H) or identical (Fig. 3I, J) mean direction became successively broken down at temperatures up to 565–600°C, and from these temperatures onwards up to 675°C (Figs. 3H–J, 4E). Both components are of reversed polarity and represent about 30 and 40% of initial NRM intensity respectively.

Arnas limestone–top Kalakot Formation Limestone specimens from the upper part of the Subathu Group were more delicate to demagnetize and to analyze than those from the Murree Group. Thermal demagnetization generally could not be pursued beyond 500°C, at which temperature

strongly viscous magnetic phases developed (Figs 3K–M, 4F, H). Initial NRM intensity ranged between 0.2 and 1.8 mA m^{-1} Four magnetic phases could be distinguished (Table 2: 21–24):

(1) A very soft component with dispersed directions and probably of viscous origin, representing up to 30% of initial NRM, was generally eliminated at 110–130°C (Fig. 3M)

(2) and (3) A characteristic combination of a normal and a reversed polarity component representing 20–40% and 15–30% respectively of initial NRM intensity, became successively broken down between 130° and 200–230°C and up to 300–320°C.

(4) A harder component, representing 10–20% of initial NRM, with predominant normal (Fig. 3K–M) and occasionally reversed polarity was broken down up to 450–500°C.

Basal Kalakot Formation Demagnetization of samples from the basal part (IKSB-F) of the Kalakot Formation was less successful (Figs. 3N–P, 4G; Table 2. 25–31). Intensities of initial NRM ranged between 0.1 and 2.3 mA m^{-1}. The magnetization of samples from 3 of the 5 sites (IKSB–D) was dominated completely by a normal polarity component (Fig 4G) of recent origin and no other magnetic component could be isolated with any degree of within-site consistency.

A more complex magnetization pattern was evident in the lowermost site IKSF (Fig. 3N). A predominant (60–70% of initial NRM) normal polarity component of soft character could be eliminated below 190–230°C, and another normal polarity but harder component (10–15% of initial NRM) became eliminated between 290° and 500°C Both components are of recent origin Another intermediate component of reversed polarity and of low intensity (10–15% of initial NRM) could be eliminated separately between 190° and 290°C

Samples from the uppermost site (IKSE, massive limestone) showed more consistent and more accurately determinable magnetic components (Figs. 3O, P, 4G, H). A soft component of appreciable intensity (20–30% of initial NRM) and of

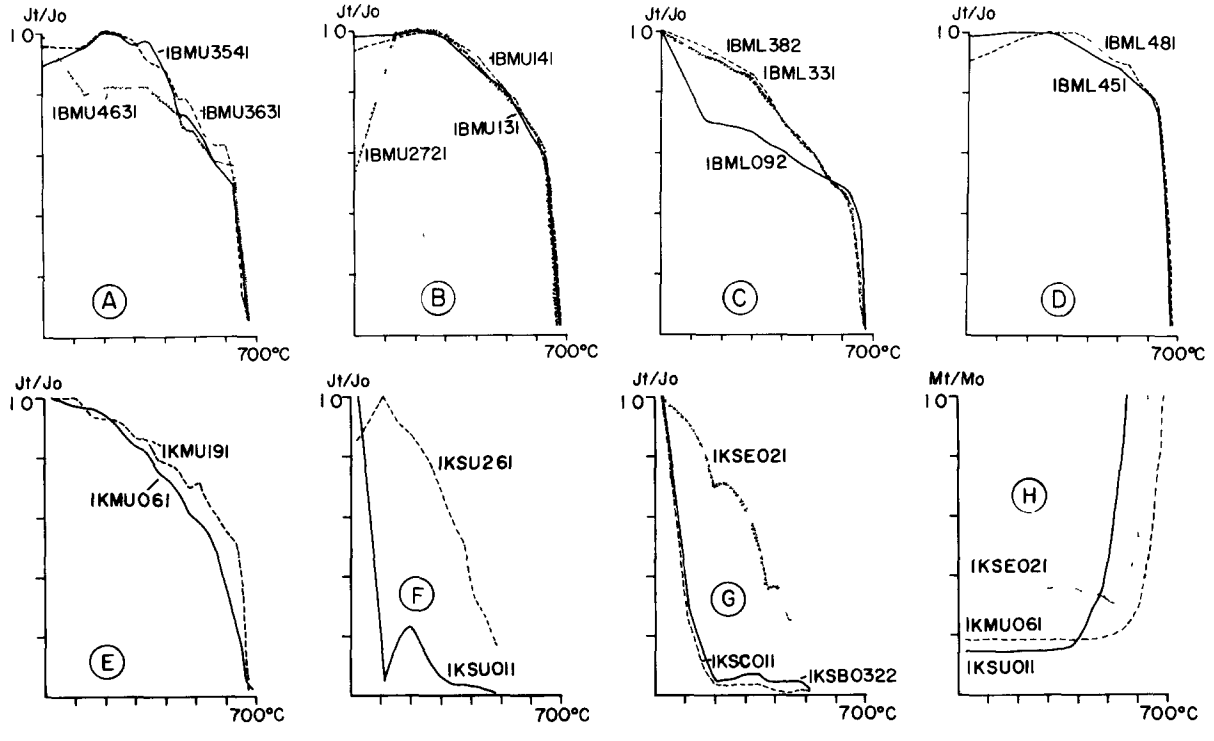


Fig 4 Representative normalized curves showing the decay of remanent magnetization (A–G) and changes in susceptibility (H) during thermal treatment

viscous origin was eliminated at 160°C. A hard component (20–40% of initial NRM) of both normal and reversed polarity was eliminated between 270° and 600°C with an intermediate component (20–40% of initial NRM) broken down between 110° and 270°C.

5. Interpretation

Directions of the soft component in the "lower upper" and "lower" Murree red beds (Fig. 3A–G: Rn) concentrate around the present field direction (Table 2: 7, 10) and the component is evidently of recent origin. The hard component with a hematite-like blocking temperature range, which shows reversed polarity in the "lower upper" Murree sediments (Figs. 3A–C, 6A) and both normal and reversed polarities in the "lower" Murree sediments (Figs. 3D–G, 6A), corresponds with palaeolatitudes of 12° and 22°N respectively (Table 2: 9, 15)

Such palaeolatitudes are to be expected in the Jammu region during Late Eocene/Early Oligocene and Late Oligocene/Early Miocene times respectively as can be concluded from internally consistent and independent sequences of palaeomagnetic poles obtained from (a) DSDP cores and palaeomagnetic data from the Indo-Pakistan continent [7,8], (b) from India-Africa relative movements fixed to a hot spot frame in the Atlantic Ocean ([19,46], Fig 5), and (c) from palaeomagnetic data from other Gondwana continents transferred to the Indian plate [9,11]. From this evidence and also on the basis of occurrence of both normal and reversed polarity directions in the "lower Murree" section we interpret this hard component to represent the primary magnetization. Its origin could be a DRM or a CRM acquired upon diagenesis. This means that our tentative field-classification based on lithological characteristics of the two stratigraphically not-interrelated sampled sections is most probably unwarranted. The palaeolatitude data suggest a late Eocene to Early Oligocene age for the "upper lower Murree" and a Late Oligocene to Early Miocene age for the "lower" Murree succession.

Interpretation of the intermediate component in the two Murree red beds successions (Fig. 3A–G) is less straightforward. The polarity change of the intermediate component in the "lower" Murree succession occurs at exactly the same level (IBML29–IBML30) as the polarity change of the hard primary component. This cannot be a mere coincidence and shows this intermediate component to be an artefact resulting from composite breakdown of both the soft and the hard magnetic component. This interpretation is the more acceptable as mean directions for both the normal and the reversed polarity component are not antiparallel, neither before nor after correction for bedding, and cannot be interpreted in any meaningful way. The direction of the intermediate component of reversed polarity in the "lower upper" Murree (Fig. 6A) can be interpreted meaningfully, though we cannot exclude the remote possibility of a similar composite nature in absence of a N/R polarity test. This component is comparable, in direction, in palaeolatitude (about 20°N, Table 2: 8), in magnetic properties, and in polarity, to intermediate components found in the basal Murree claystone and in the Subathu Group limestone (Table 2, Fig. 6B–D) Likewise it is interpreted to be of secondary origin. All these components show palaeolatitudes of origin comparable to secondary components (Fig. 7) observed in the Kashmir Basin [35], the Krol Belt [47], and possibly in the Lhasa region of southern Tibet [9,19,48–51]. These components originated during the Middle Tertiary, as is clear from the intersection of the swath formed by the loci of these secondary pole positions with the Indian APWP (Fig. 7). A possible origin for these Middle Tertiary components has been discussed before [9,10] in terms of a thermo-chemical event associated with formation of the MCT, possibly enhanced by later outwards thrusting of the more external Himalayan tectonic belts. This interpretation is extended to the here observed intermediate components.

The soft normal polarity and the intermediate reversed polarity component (palaeolatitude 23 5°N) in the basal Murree claystone (Table 2: 16, 17) are interpreted likewise as components of recent origin and of an MCT- and thrust-related origin respectively. The two hard components of closely comparable direction but with magnetite-like and hematite-like blocking temperature ranges (Fig. 6D· 18, 19) are interpreted as a primary DRM (Fig. 6D: 18, palaeolatitude 5°N) and a component acquired shortly afterwards upon diagenesis of the beds (Fig. 6D. 19, palaeolatitude

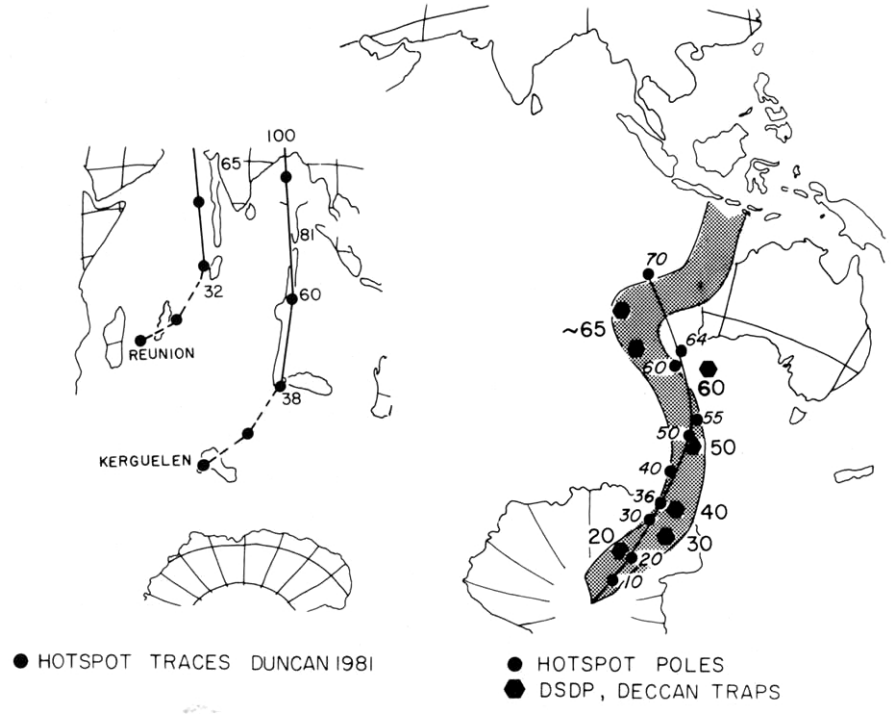


Fig 5 The lefthand figure indicates hotspot traces for the Reunion and Kerguelen hotspots, predicted from analysis of hotspot traces, in the Atlantic Ocean combined with India-Africa relative movement data [46] Full dots indicate predicted positions at about 20 Ma intervals The righthand figure shows a comparison of Indian Tertiary palaeomagnetic poles simulated from the above India-Africa relative movement data fixed to a hotspot frame in the Atlantic Ocean (full dots, italics indicate ages in Ma), with palaeomagnetic poles simulated from palaeolatitude observations from DSDP cores on the Indian plate and with palaeomagnetic poles from the Deccan Traps [20,21]

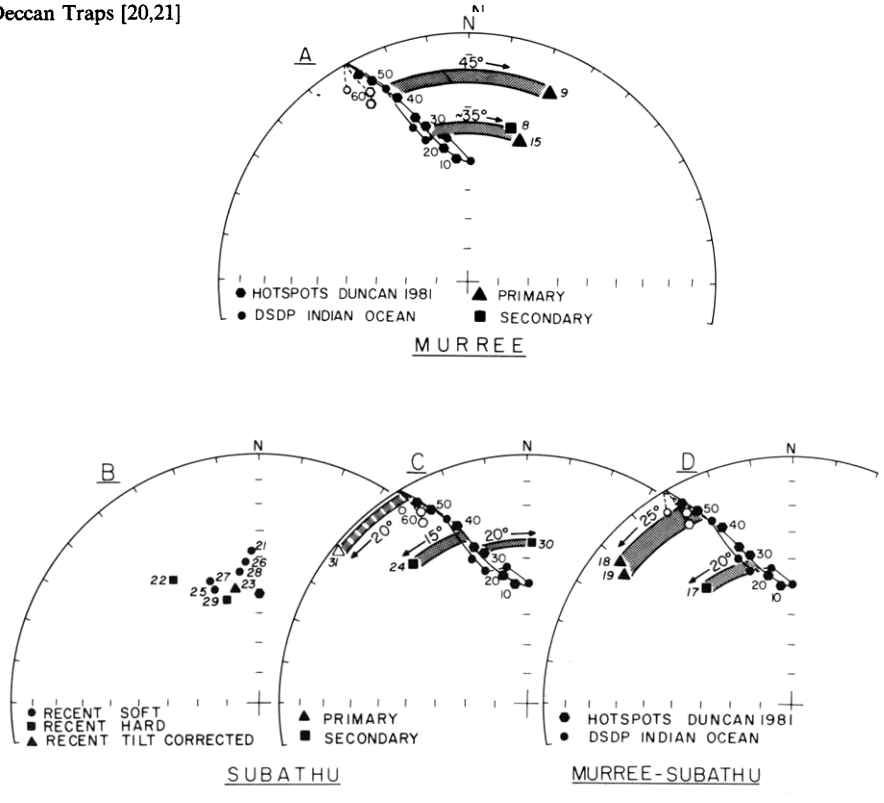


Fig 6 Comparison of directions observed for the primary, secondary (intermediate) and recent field components with directions expected in the sampled region according to the simulated Tertiary APWP's, see Fig 5 Italic numbers correspond with results listed in Table 2 The other numbers indicate ages in Ma or the magnitude of rotation relative to the Indian shield

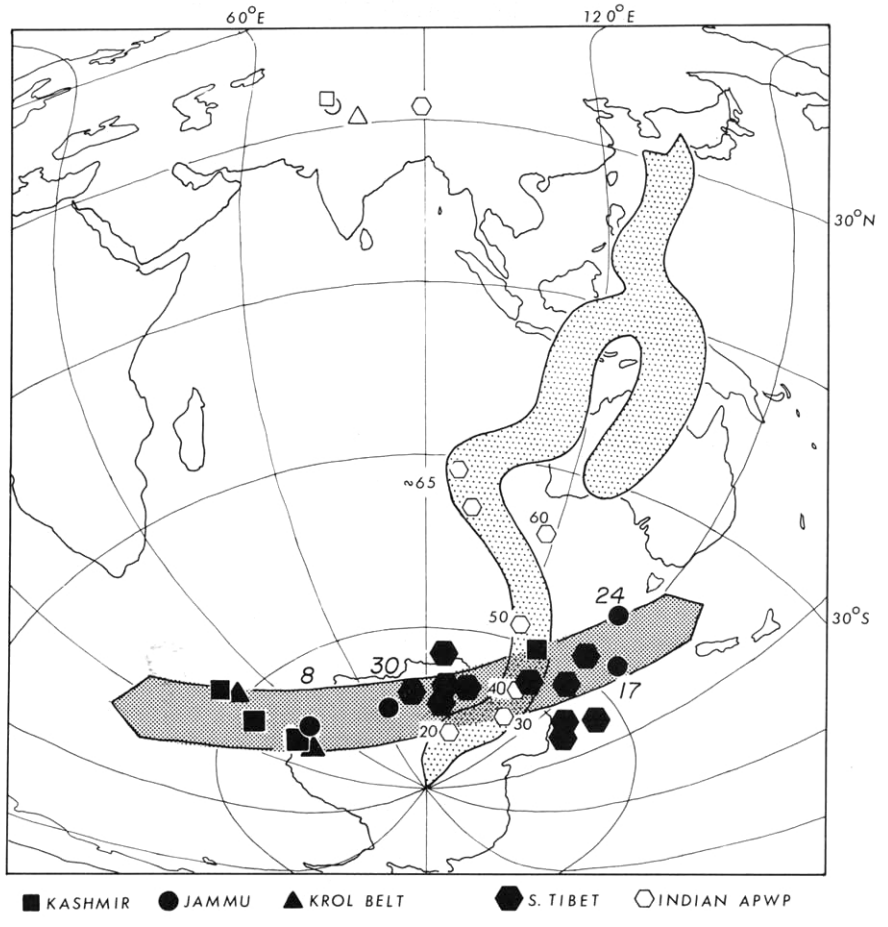


Fig 7 Comparison of the Indian APWP since the Late Palaeozoic with a swath formed by loci of pole positions for secondary components observed in the present study, in Kashmir [35], and in the Krol Belt [47] Several pole positions from Middle Cretaceous and Lower Tertiary formations from the Lhasa block for which a similar secondary origin cannot be excluded [48–51] are shown as well Pole positions are not corrected for rotations relative to the Indian shield Italic numbers indicate ages in Ma, the other numbers correspond with results listed in Table 2 Aitoff projection

7 5°N) respectively (compare Tauxe and Badgley [52], who describe various phases of remanence acquisition in hematite occurring during and immediately after deposition for the younger Siwalik red beds from the Potwar region (Pakistan)). Though determined with appreciable accuracy (α_{95} values of 4.5° and 3.5° respectively), their directions are not significantly different at the 95% probability level and a mean direction based on individual specimen directions computed over the combined blocking temperature intervals of both components (400–675°C) is proposed as the best estimate for the primary magnetization of these rocks (Table 2· 20).

The striking combination of a soft normal polarity and an intermediate reversed polarity component in the upper part of the Subathu Group (Figs. 3K–M, 6B; Table 2 21, 23) represents recent field components of post-folding (N-polarity: 21) and pre-folding (R-polarity: 23) origin respectively. Harder components of predominantly normal polarity show two groups of directions, corresponding either with a post-folding recent field component (Fig. 6B, Table 2· 22) or with a palaeolatitude of 15°N (Figs. 6C, 7, Table 2: 22) Primary components in the overlying basal Murree claystone and in the underlying basal part of the Kalakot Formation correspond with palaeolatitudes of 6.5°N and 1°S respectively There is an obvious difficulty in interpreting this

15°N component as a primary one, because of seafloor spreading evidence for a steady northward movement of the Indian plate. A secondary origin, possibly associated with formation of the MCT as observed throughout this study and elsewhere, seems more probable.

The samples taken from the basal part of the Kalakot Formation (sites IKSB–F) showed predominantly recent field components residing in magnetic fractions with both low and high blocking temperature spectra (Fig. 6B, Table 2: 25–28, 29). Notable exceptions are site IKSF with an intermediate component whose corresponding palaeolatitude of 19°N (Fig. 6C; Table 2 30) suggests a secondary MCT related origin (Fig. 7), and site IKSE with a hard component of predominantly reversed polarity (Fig. 6C, Table 2: 31) whose corresponding palaeolatitude of 1°S indicates a primary origin. Interpretation of the latter component is confirmed by recent findings of primary components of comparable equatorial palaeolatitude in carbonates and sandstones of latest Palaeocene to earliest Eocene age from southern Tibet [53].

6. Discussion

Detailed knowledge of the Tertiary segment of the Indian APWP is a prerequisite to interpret palaeomagnetic results from the Himalayan region in terms of post-collisional deformation. Efforts to detail this Tertiary APWP directly from continental palaeomagnetic data, have not met with success sofar (Klootwijk, unpublished results, Cutch and Jaisalmer regions) Simulated APWP trajectories, derived from various sources (Fig. 5), are in fair agreement and can be taken as reliable estimates for the Indian Tertiary APWP. Comparison of directions expected in the Riasi thrust sheet in the Jammu foothills according to these APWP segments [7,8,19,46] and directions actually observed, indicate large-scale rotations with respect to the Indian shield.

Declinations for the primary and the secondary components in both the "lower upper" and the "lower" Murree succession are distinctly offset in a clockwise sense with respect to the expected pattern (Figs. 6A, 8). The primary component from the Upper Eocene/Lower Oligocene "lower upper" Murree succession shows an about 45° offset. The secondary component from this succession and the primary component from the Upper Oligocene/Lower Miocene "lower" Murree succession show a smaller about 35° offset. This difference suggests a minor rotation during the Middle Tertiary, which has been concluded also for the Krol Belt [47].

These rotations are strikingly similar in sense and magnitude to rotations observed for other thrust units throughout the northwestern Himalaya (Figs. 8, 9), i e. the Panjal Nappe (45° [35]), the Krol Belt (45° [47]), and to some extent the Thakkhola region of western Nepal (10–15° [54]).

Whilst we cannot exclude the possibility of surprisingly coinciding patterns of local rotations around nearby Eulerian poles, the coherency of the rotations and the continuity along strike of the tectonic units strongly point to large-scale rotational overthrusting of coherent tectonic units. The Eulerian pole of rotation must be situated close to the northwestern extreme [54] of the IKSZ, the subsurface continuation of the MCT-MBT beyond the syntaxial region [34] This simple model [19] of an eastwards increasing magnitude of rotational overthrusting onto the Indian shield, gives magnitudes of 550 and 650 km for the Krol Belt and the Thakkhola region respectively (Fig. 9). Balanced cross sections over the Western Himalayan Syntaxis and the Pamirs estimate the magnitude of crustal shortening in that region, i.e. Greater India's underthrusting, to be in excess of 470 km [55] and 600 km [56] respectively. Our model [19] accounts for a negligible or minimal magnitude of rotational under(over)thrusting in that region, as it is close to the assumed Eulerian rotation pole. Clearly, crustal shortening modelled by rotation around a distant pole plays an important role in the Himalaya. Its magnitude may be determined along the arc from carefully constructed balanced cross sections. The above estimates for rotational overthrusting of 550 and 650 km respectively place no more than a lower limit on the magnitude of crustal shortening and we see this as strong support for Powell and Conaghan's [13,17] and Veevers' [14] model of an original northern extent of Greater India reaching at least to the southern outline of the Kun Lun in its present-day position with respect to the Indian shield

Good agreement between the Early Eocene

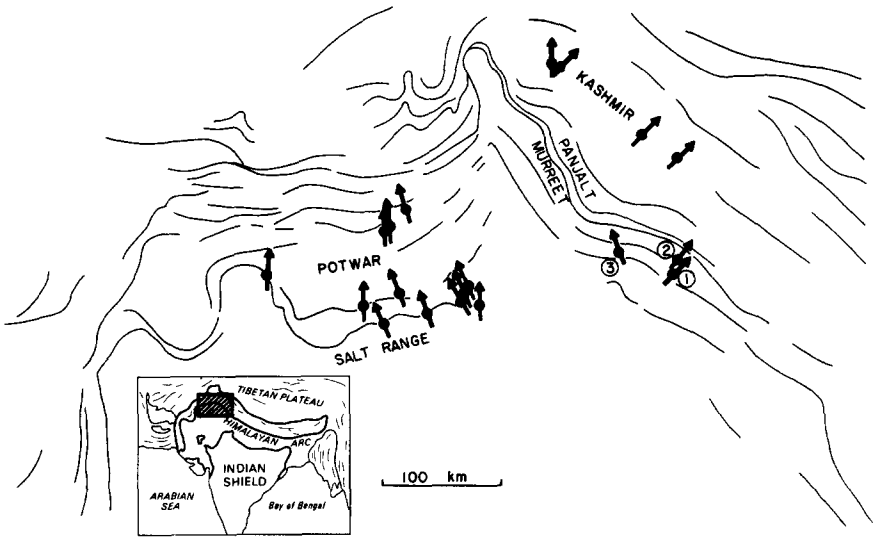


Fig 8 Comparison of structural trends in the syntaxial region after Calkins et al [64] with palaeomagnetic rotation vectors whose azimuth with respect to north indicates the magnitude of rotation of the studied tectonic unit with respect to the Indian shield *1* = "lower upper" Murree succession, *2* = "lower" Murree succession, *3* = Kalakot inlier Indicated also are rotation vectors observed for the Kashmir region [35] and for the Potwar basin and the Salt Range [1,9] Compare Fig 1 for location

palaeolatitude observed for Tingri [53], north of the MCT, and the palaeolatitude expected at that location within a reconstructed Greater India, corrected for the here determined magnitudes of rotational overthrusting, is shown in Fig. 10.

The above observations and the consistent per-

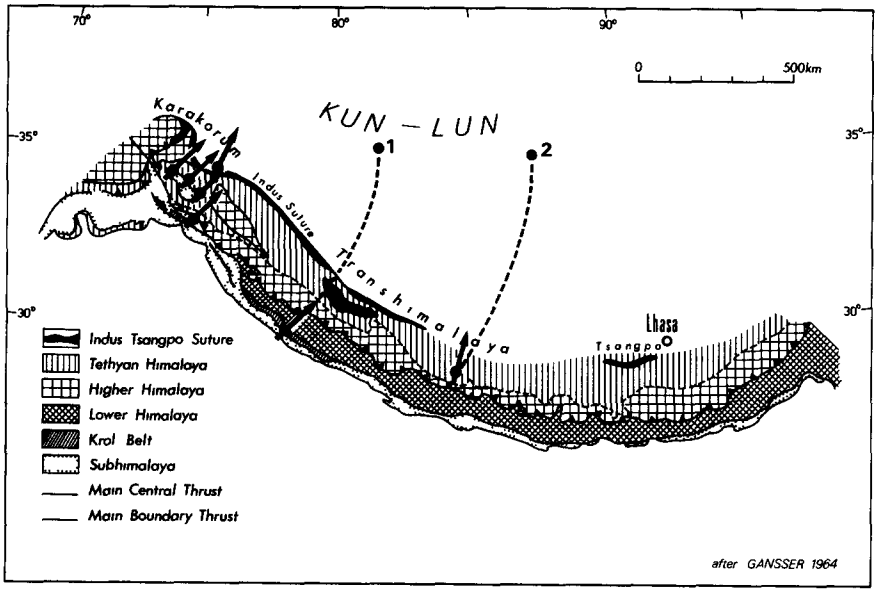


Fig 9 Schematic map of the Himalayan Arc after Gansser [63] with "palaeomagnetic rotation vectors" observed in the present study, and for the Kashmir regions [35], Ladakh [58,65], the Krol Belt [47] and the Thakkhola region [54] The two full dots indicate the position of the Krol Belt (*1*) and the Thakkhola (*2*) region respectively, when relocated with respect to the Indian shield after correction for a coherent rotation around an Eulerian pole situated at the northwestern extreme of the Indus Kohistan Seismic Zone [34] The magnitude of this relocation is taken in first approximation as an uniform 45° clockwise rotation up to the Krol Belt and an uniform 15° clockwise rotation beyond up to the Thakkhola region Note that the relocated positions fall close to the Kun Lun This northward relocation does not account for crustal shortening over the Himalayan Belt as observed from balanced cross sections [55,56] and has to be considered as a minimal estimate only

pendicularity, where measured, along the Himalayan Arc between the "palaeomagnetic rotation vector" and the main structural trend (Figs. 8, 9), are discussed in a model linking oroclinal bending and a conical downwarp in the Great Vindhyan and Ganges Basin of the underthrusting Indian shield with the extensional regime in south central Tibet [19].

The palaeomagnetic results for the Kalakot region form a notable but understandable exception to the gradual variation of the "palaeomagnetic rotation vector" along the Himalayan Arc [19]. Declinations from the primary and secondary components of the basal Murree claystone and the Subathu Group (Fig. 6C, D, Table 2. 17–19, 24, 31) show an aberrant counterclockwise rotation over 20–25° (except site IKSF, Fig. 6C; Table 2: 30) with respect to the Indian shield. This rotation is at least partly of recent origin as indicated by a counterclockwise offset of the recent field components (Fig 6B, Table 2: 21–23, 25–29). The general NW-SE structural trend of the northwestern Himalaya deviates in the Kalakot region to an east-west trend [57] and the perpendicular orientation of the "palaeomagnetic rotation vector" and the regional trend still holds for this region (Figs 1, 8). Clearly, this aberrant rotation results from a local flexure in structural trend similar to the local absence of rotation in northwestern Kashmir close to the syntaxial region where structural trends also deviate to an east-west direction (Fig. 8) [35].

Palaeolatitudinal positions of Greater India according to the results from the basal Murree claystone and the basal beds of the Kalakot Formation (Fig. 10: 1, 2) are in good agreement with positions expected according to the simulated Tertiary APWP for India (Fig. 10). These data show that Greater India, in its here documented minimal northern extent (Fig. 9) [13,14,17,19], crossed between about 60 Ma and about 55 Ma ago the

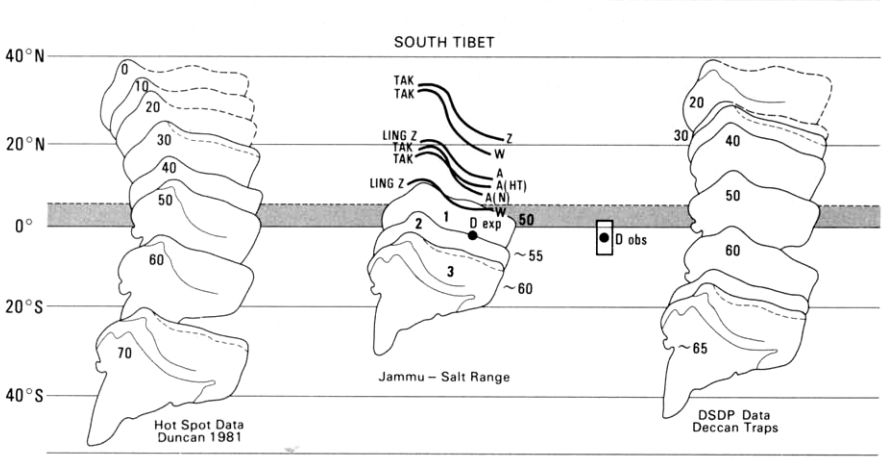


Fig 10 Sequential positions of Greater India according to palaeomagnetic pole positions simulated from India-Africa relative motion data fixed to a hotspot frame in the Atlantic Ocean (left [46]), palaeolatitude observations from DSDP cores on the Indian plate and palaeomagnetic data from the Deccan Traps (right [7,8]), and results obtained in this and in the accompanying paper ([1], center) See also Fig 5 Less pronounced numbers indicate ages in Ma *1* = basal Murree beds (Table 2 20), *2* = Kalakot Formation (Table 2 31), *3* = Lockhart Limestone [1] The four outlines indicated in Greater India represent from south to north the Himalayan mountain front along the Indo-Gangetic basin, the Indus-Tsangpo Suture zone (ITS), the minimal northern extent of Greater India according to the reconstruction described, which falls close to the southern outline of the Kun Lun–Astin Tagh–Nan Shan, and the same outline with the choked-off Tethyan Himalaya added to it The outline for southern Tibet (center) is taken as the present-day outline of the ITS The palaeoposition of this zone is indicated according to results from the Middle Cretaceous Takena Formation (Tak Z = Zhu Zhi Wen et al [51], W = Westphal et al [48], A = Achache et al [49, N and HT are hard magnetic components observed in the Lhasa Block to the north and to the south of the Nyaiqentanglha Range respectively]), and the Lower Tertiary Lingzizong Formation (Ling Z A = Achache [49], W = Westphal et al [48]) D_{obs} and D_{exp} indicate a very good agreement between the observed latest Palaeocene/earliest Eocene palaeolatitude in Tingri, southern Tibet ([53], with 95% confidence limits), and the palaeolatitude expected within a reconstructed Greater India according to result 2 for the Kalakot Formation (Table 2 31) The shaded belt between 0° and 7°N represents the palaeolatitude of origin for collision-related secondary magnetic components observed in the Eastern Hindu Kush [9,19,59], Ladakh [58,65] and the Thakkhola region [54] Mercator projection

equatorial to 7°N belt. This belt has been interpreted from collision related secondary palaeomagnetic data observed north and south of the ITS [9,19,58,59] as the zone of initial contact and suturing between Greater India and south-central Asia.

Progressive suturing from west to east along the ITS has been concluded [60,61] from more prolonged generation of subduction related magma in the Lhasa region (until about 40 Ma) than in Ladakh (until about 60 Ma) This accords with seafloor spreading evidence for a suturing related phase of counterclockwise rotation of the Indian plate during the Palaeocene [19,62] and with the palaeomagnetic results from the present and accompanying papers [1,19]. The latter data suggest that initial contact of northwestern Greater India had been established before about 60 Ma ago in the Hindu Kush–Karakorum region (Fig. 10: 3) and around 55 Ma ago in the Lhasa region.

Comparison of palaeomagnetic data from India and from the Lhasa Block of southern Tibet [48–51] shows a discrepancy. The majority of results from the Albian/Aptian Takena Formation and from the Early Tertiary (60–48 Ma) Lingzizong Formation of the Lhasa Block show far higher northern palaeolatitudes for Tibet's southern boundary than for Greater India's reconstructed northern boundary at the time of suturing (Fig. 10) The possibility of a Middle Tertiary remagnetization within the Lhasa Block [9] similar to that observed in the more external Himalayan thrust sheets (this paper, [9,10]) was refuted by the French-Chinese authors on grounds of a positive fold test [48–50]. It cannot be excluded, however, that the outcome of such fold tests are biased, as the effect of post-Lingzizong deformation (later than 60–48 Ma) is inadequately known.

7. Conclusions

Palaeomagnetic results from the Tertiary succession of the Jammu foothills further establishes the widespread occurrence in the more external Himalayan belts of a Middle Tertiary magnetic overprint which is associated with formation of the MCT and with outwards thrusting of the Lower Himalayan tectonic belts. The 45° clockwise rotation relative to the Indian shield observed in the Murree red beds succession northeast of Jammu accords with similar rotations in the Panjal Nappe, the Krol Belt, and lesser rotations in the Thakkhola region. These data strongly suggest that at least the northwestern and the central Himalayan region have overthrusted the Indian shield with a coherent large-scale clockwise rotation whose magnitude amounts to minimal 550 km in the Krol Belt and minimal 650 km in the Thakkhola region These minimal and certainly underestimated magnitudes of crustal shortening strongly support Powell and Conaghan's [13,17] and Veevers et al.'s [14] model for Greater India and large-scale intracontinental underthrusting along the MCT. Comparison of the Indian APWP and palaeolatitude observations from the Jammu foothills and the Trans-Indus Salt Range [1] with palaeomagnetic data from the Lhasa Block support isotope geochemical evidence from southern Tibet's pre-collisional Andean margin for west to east progressive suturing of the ITS. The data show, however, higher northern palaeolatitudes for the southern margin of the Lhasa Block at the time of suturing than expected from Indian palaeomagnetic data.

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